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Course Code: ITA0603

Course Name: Machine Learning

Course Assignment, Deliverables & Evaluation Framework

1. Course Metadata & Institutional Framework Alignment

Course Code & Name:	ITA0603 – Machine Learning
Assignment Title:	Development and Empirical Evaluation of High-Scale Predictive Maintenance and Asset Optimization Infrastructure using Hybrid Machine Learning Paradigms
Bloom's Taxonomy Levels:	L4 – Analyse; L5 – Evaluate; L6 – Create
SDG Mapping:	SDG 9 – Industry, Innovation, and Infrastructure (Specifically Target 9.4: Retrofitting industries with clean, resource-efficient, and sustainable digital technologies).

Course Outcome (CO) Mapping Matrix

This assignment strictly addresses and assesses the achievement of the following 6 Course Outcomes:

- **CO1:** Acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search.
- **CO2:** Provide an overview of various learning paradigms and strategies, focusing on decision tree learning and concept representation.
- **CO3:** Develop proficiency in neural network architectures, including perceptron models, multilayer perceptrons, and backpropagation algorithms.
- **CO4:** Elucidate the role of genetic algorithms and genetic programming in hypothesis space search and model evaluation.
- **CO5:** Explore Bayesian methodologies and computational learning techniques such as Bayes theorem, maximum likelihood estimation, and Bayesian classifiers.
- **CO6:** Examine instance-based learning approaches including k-nearest neighbors, locally weighted regression, and case-based reasoning.

2. Integrated Core Problem Statement

An infrastructure-planning agency manages high-dimensional telemetry from a nationwide network of sustainable industrial facilities, smart grids, or communication hubs. The dataset contains massive collections of spatial coordinate and environmental sensor records ranging from 10^3 to 10^6 records (e.g., global cell towers or weather station sensory networks). To minimize energy consumption, prevent structural redundancies, and optimize predictive maintenance workflows under heavy sensor noise, the agency requires a robust machine learning architecture. Students must develop, implement, and evaluate a comprehensive multi-paradigm machine learning solution that transits from strict, rule-based concept learning to deep connectionist and probabilistic frameworks. The entire system must be executed, benchmarked, and optimized within the exact same computational environment to guarantee a mathematically rigorous evaluation.

3. Detailed Assignment Task Breakdowns

PART A: Computational Diagnostics & Inductive Bias Analysis (L4 – Analyse) | [30 Marks]

- **1. Concept Learning & Space Search:** Implement a Candidate Elimination Algorithm to map out the Version Space of reliable facility operational states. Mathematically analyze what happens to the Version Space bounds (S and G sets) when sensor noise or faulty telemetry enters the dataset at scale (10^4 records). Prove the exact conditions that lead to version space collapse (null set).
- **2. Heuristic Space Search vs. Restriction Bias:** Implement an ID3/C4.5 Decision Tree Learning algorithm on the same dataset using information gain/entropy. Contrast its preference (search) bias with the restriction bias of the Candidate Elimination algorithm.
- **3. Mathematical Complexity:** Using Big-O, Omega, and Theta notations, analyze the theoretical space and time complexity of searching these hypothesis spaces as the scale of the infrastructure records increases from 10^3 to 10^5 variables.

PART B: Connectionist vs. Probabilistic Architecture Evaluation (L5 – Evaluate) | [35 Marks]

- **1. Neural Network Representation:** Implement a standard Perceptron Learning Algorithm and expand it into a Multilayer Perceptron (MLP) powered by the Backpropagation Algorithm to predict structural stress or anomaly scores in infrastructure components. Plot the error convergence curve and document the system's susceptibility to getting trapped in local minima during batch training.
- **2. Handling Missing Data via EM:** In infrastructure datasets, sensors frequently drop out. Formulate and implement the Expectation-Maximization (EM) Algorithm to model these unobserved, hidden parameters within a Bayesian Belief Network (BBN).
- **3. Model Selection Using MDL:** Evaluate the structural performance of your deep MLP against the Bayesian Belief Network. Use the Minimum Description Length (MDL) Principle to mathematically score both models, balancing training error against architectural complexity: $\text{MDL Cost} = -\log_2 P(D|h) + (|h|/2) \log_2 |D|$. Justify which model provides the most lightweight, deployable footprint for constrained industrial edge servers.

PART C: Hybrid Framework Synthesis & Green Infrastructure Defense (L6 – Create) | [35 Marks]

- **1. Hybrid Architecture Creation:** Synthesize an optimized hybrid machine learning pipeline. Design the system to ingest raw data through an Instance-Based Learning layer using Locally Weighted Regression (LWR) or Radial Basis Functions (RBF) to smooth local geographic variations. Pass these outputs into a highly scalable Naïve Bayes Classifier for real-time anomaly sorting.
- **2. Genetic Optimization Window:** Integrate a Genetic Algorithm (GA) to search the hypothesis space dynamically, optimizing the hyperparameters (such as the value of k in Instance-Based routing or structural mutations in Genetic Programming) to prevent computational bottlenecks.
- **3. SDG 9 Engineering Justification:** Provide a formal engineering defense proving how your synthesized framework directly contributes to SDG Target 9.4. Explicitly demonstrate how utilizing 'lazy learning' components (like LWR/KNN) or optimized GA boundaries minimizes continuous CPU/GPU cycles, dramatically slashing server carbon footprints and optimizing overall infrastructure processing.

4. Mandatory Student Deliverables

Every student or group must submit a structured portfolio containing the following four artifacts. Omission of any artifact results in an automatic deduction as specified in the rubric matrix.

- **Deliverable 1: Source Code Repository & Execution Pipeline.** A clean, executable, well-commented source code codebase (Python 3.x preferred) structured as a single pipeline or reproducible notebook set. The script must run the algorithms over the scaled benchmarks (10^3 to 10^4 data points) in a unified testing sequence.
- **Deliverable 2: Mathematical Derivations & Analytical Notebook.** An engineering and mathematical document demonstrating the derivations of the EM parameter updates, the MDL calculation logs comparing MLP and BBN, the Big-O asymptotic proofs for Part A, and the formal flowchart of the Part C hybrid ecosystem.
- **Deliverable 3: Empirical Benchmarking & Visualization Report.** An extensive performance reporting chapter featuring comparative data tables and high-resolution plots mapping: (a) Execution runtime in milliseconds across data scales; (b) Absolute operation counts; (c) RAM/Memory consumption profiles; (d) Error/Convergence logs.
- **Deliverable 4: Sustainable Infrastructure Technical Defense Paper.** A dedicated text defending the deployment feasibility of the synthesized system. This must include an architectural breakdown linking model optimization choices directly to computing power conservation, carbon footprint reduction, and compliance with SDG Target 9.4.

5. Comprehensive 4-Tier Evaluation Rubrics Matrix

Grading will be strictly measured using the following multi-dimensional rubric grid (Total: 100 Marks):

Evaluation Criteria	Excellent (85-100%)	Proficient (70-84%)	Developing (50-69%)	Unsatisfactory (<50%)
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Criteria Mapping) (CO	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Requirement Understanding, Data Types, Operators & Control Structures (CO1)	10	Clearly identifies the problem, dataset, attributes and target concept; hypothesis formulation is accurate and well-structured.	Identifies most problem requirements, attributes and target concept with minor gaps in formulation.	Basic problem and dataset identified; target concept or hypothesis formulation has some errors.	Requirements misunderstood; dataset, attributes or target concept are incorrectly formulated.
Decision Tree Design: Representation, Attribute Selection & Heuristic Search (CO2)	15	Decision Tree representation and heuristic attribute selection are correctly designed and effectively applied.	Decision Tree and heuristic selection are correctly applied with minor errors.	Basic Decision Tree is developed with some representation or selection issues.	Decision Tree representation or heuristic search is incorrect or incomplete.
Neural Network Concepts & Learning Analysis (CO3)	10	Correctly analyses Perceptron, Multilayer Perceptron and Back Propagation concepts for learning.	Analyses neural-network concepts with minor gaps or errors.	Demonstrates basic understanding of neural-network learning.	Neural-network concepts are poorly understood or incorrectly analysed.
Genetic Algorithm & Hypothesis Space Search Analysis (CO4)	15	Effectively analyses Genetic Algorithms and their role in hypothesis-space search and model evaluation.	Correctly explains GA-based search with minor gaps.	Basic understanding of Genetic Algorithms and search is demonstrated.	Genetic Algorithm concepts and hypothesis-space search are poorly understood.
Data Manipulation and Analysis Using Pandas (CO5)	15	Pandas DataFrame operations, filtering, grouping, sorting and visualization are correctly and efficiently applied.	Pandas operations are correctly applied with minor errors or gaps.	Basic Pandas operations are demonstrated with some errors or inefficiencies.	Pandas operations are missing, incorrect or poorly implemented.

Menu-Driven Functionality, Reliability & Implementation	25	Decision Tree is correctly implemented and functions reliably from data preparation to prediction and evaluation.	Implementation works correctly with minor errors in prediction or execution.	Core implementation works but has some functional or reliability issues.	Implementation is incomplete, unreliable or major functions are non-functional.
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Thoughtful justification of design choices, clear connection to SDG 7/9/12/13 relevance, and honest, specific account of challenges faced and learning gained.	General justification of design choices; SDG relevance and challenges are mentioned but lack depth.	Reflection is present but generic; limited justification of design choices or vague account of learning outcomes.	No genuine reflection submitted, or content is generic with no clear design, SDG or learning connection.

Institutional Framework Note: This framework is customized for academic evaluation purposes. Ensure identical computational environments are strictly provisioned during lab sessions to achieve matching empirical baselines.

ANSWERS

1.Introduction

The proposed system implements a multi-paradigm machine learning framework for sustainable infrastructure monitoring. The framework combines symbolic learning, connectionist learning, probabilistic learning, instance-based learning, and evolutionary optimization.

The system processes infrastructure sensor data and performs anomaly detection while evaluating computational efficiency. The implementation includes Candidate Elimination, ID3, Perceptron, Multilayer Perceptron, Expectation-Maximization, Bayesian Belief Network, Minimum Description Length, Locally Weighted Regression, Naïve Bayes, and Genetic Algorithm techniques.

All algorithms are executed in the same Python environment to provide a consistent comparison of runtime, memory usage, prediction accuracy, and convergence.

2. Software and Hardware Requirements

Software Requirements

- Python 3.x
- NumPy
- Pandas
- Matplotlib
- Scikit-learn
- Tracemalloc

Hardware Requirements

- Standard laptop/desktop computer
- Minimum 8 GB RAM recommended
- Multi-core processor recommended

3. Dataset Description

A synthetic infrastructure sensor dataset is generated to represent sustainable industrial facilities, smart grids, and communication infrastructure.

The dataset contains the following attributes:

Attribute	Description
Latitude	Geographic latitude of the facility
Longitude	Geographic longitude of the facility
Temperature	Environmental temperature
Humidity	Environmental humidity
Pressure	Atmospheric pressure
Vibration	Structural vibration level
Noise	Environmental/sensor noise
Stress	Structural stress level
Power	Power consumption
Wind Speed	Wind condition
Anomaly	Target class indicating normal or anomalous condition

The dataset is generated with controlled random noise so that the experiments are reproducible.

4. Data Preprocessing

The generated sensor data is processed before machine learning.

The preprocessing steps are:

1. Generate infrastructure sensor data.
2. Separate input features and target values.
3. Standardize numerical features.
4. Divide the dataset into training and testing sets.
5. Introduce missing values for the EM experiment.
6. Convert continuous attributes into categorical values for Candidate Elimination and ID3.

Standardization is performed using:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original value, μ is the mean, and σ is the standard deviation.

5. Part A – Computational Diagnostics and Inductive Bias Analysis

5.1 Candidate Elimination

Candidate Elimination is used to determine the Version Space of hypotheses that are consistent with the training examples.

The algorithm maintains two boundaries:

- Specific boundary S
- General boundary G

The Version Space is defined as:

$$VS(H, D) = \{h \in H : h \text{ is consistent with } D\}$$

If no hypothesis is consistent with the complete dataset, the Version Space becomes:

$$VS(H, D) = \emptyset$$

This represents Version Space collapse.

The implementation evaluates the effect of noisy or conflicting sensor observations on the hypothesis boundaries.

5.2 ID3 Decision Tree

The ID3 algorithm constructs a decision tree using entropy and information gain.

Entropy is calculated using:

$$H(S) = - \sum_i p_i \log_2(p_i)$$

Information gain is calculated using:

$$Gain(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v)$$

The attribute having the highest information gain is selected as the decision node.

Candidate Elimination primarily demonstrates restriction bias, whereas ID3 demonstrates a preference/search bias by selecting attributes that provide maximum information gain.

5.3 Complexity Analysis

The algorithms are evaluated for increasing dataset sizes.

The experimental analysis considers:

- Execution time
- Memory consumption
- Computational operations
- Accuracy

Theoretical complexity is expressed using:

$$O(f(n))$$

$$\Omega(f(n))$$

$$\Theta(f(n))$$

where n represents the number of records.

6. Part B – Connectionist and Probabilistic Architecture

6.1 Perceptron

The Perceptron is implemented as a basic single-layer neural classifier.

The prediction is represented as:

$$\hat{y} = f(w^T x + b)$$

The weights are updated using:

$$w_{new} = w + \eta(y - \hat{y})x$$

The Perceptron provides a baseline for comparison with the more advanced MLP model.

6.2 Multilayer Perceptron

A Multilayer Perceptron is implemented using multiple neural layers.

The model performs:

1. Forward propagation.
2. Loss calculation.
3. Backpropagation.
4. Weight update.
5. Repeated training until convergence or the maximum number of epochs.

The error convergence curve is generated to study the learning behavior of the network.

The training process is also discussed in terms of local minima, saddle points, and flat optimization regions.

6.3 Expectation-Maximization

The Expectation-Maximization algorithm is used to estimate missing or hidden sensor information.

The algorithm consists of two stages.

E-Step

The hidden variables are estimated using the current parameter values.

$$Q(\theta \mid \theta^{old}) = E[\log P(X, Z \mid \theta)]$$

M-Step

The model parameters are updated by maximizing the expected log-likelihood.

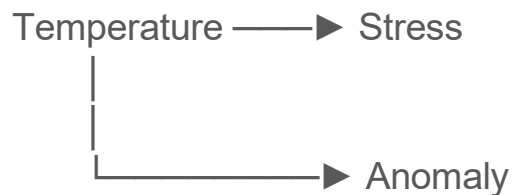
$$\theta^{new} = \arg \max_{\theta} Q(\theta \mid \theta^{old})$$

The process is repeated until the parameter change becomes smaller than the selected convergence threshold.

6.4 Bayesian Belief Network

A Bayesian Belief Network is used to represent probabilistic relationships between infrastructure variables.

For example:



Vibration ———▶ Stress

The network estimates conditional probabilities and uses them to determine the likelihood of an infrastructure anomaly.

6.5 Minimum Description Length

The Minimum Description Length principle is used to compare the MLP and BBN models.

The MDL cost is:

$$MDL = -\log_2 P(D \mid h) + \frac{|h|}{2} \log_2 |D|$$

The model with a lower MDL score provides a better balance between

predictive performance and model complexity.

For edge deployment, MDL is considered together with:

- Runtime
- Memory consumption
- Number of parameters
- Prediction accuracy

7. Part C – Hybrid Framework

7.1 Locally Weighted Regression

Locally Weighted Regression is used to smooth local geographic and sensor variations.

The weight assigned to each training observation is:

$$w_i = e^{-\frac{\|x_i - x_q\|^2}{2\tau^2}}$$

where:

- x_i is a training observation.
- x_q is the query point.
- τ is the bandwidth parameter.

This allows nearby observations to have greater influence than distant observations.

7.2 Naïve Bayes

The processed data is passed to a Gaussian Naïve Bayes classifier.

The classification probability is:

$$P(C | X) \propto P(C) \prod_i P(x_i | C)$$

The classifier predicts the class with the highest posterior probability.

The possible outputs are:

- Normal
- Anomaly

7.3 Genetic Algorithm

A Genetic Algorithm is used to optimize the parameters of the hybrid system.

The main operations are:

1. Population initialization.
2. Fitness evaluation.

3. Selection.
4. Crossover.
5. Mutation.
6. Generation of a new population.
7. Selection of the best solution.

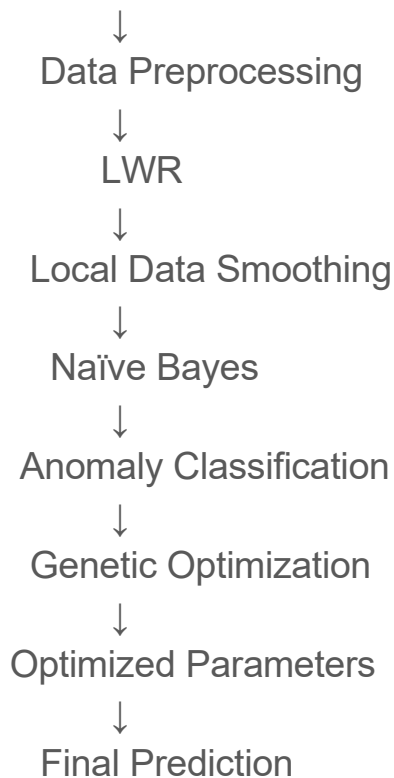
The fitness function combines predictive performance and computational efficiency:

$$Fitness = Accuracy - \lambda(Runtime) - \mu(Complexity)$$

The optimized parameter is then used by the final hybrid model.

7.4 Final Hybrid Framework

Raw Infrastructure Sensor Data



8. Experimental Benchmarking

All algorithms are executed under the same computational environment.

The following parameters are measured:

Runtime

$$Runtime = EndTime - StartTime$$

Memory

Peak memory consumption is measured during algorithm execution.

Accuracy

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Error

$$Error = Actual - Predicted$$

The experiments are repeated for increasing dataset sizes.

The benchmark results are stored automatically in a CSV file.

9. Results and Visualizations

The implementation generates the following results:

1. Dataset size versus execution time.
2. Dataset size versus memory consumption.
3. Dataset size versus prediction accuracy.
4. MLP epoch versus training loss.
5. EM convergence results.
6. MLP and BBN MDL comparison.
7. Genetic Algorithm optimization results.

The generated plots are saved automatically in the results/figures directory.

10. Conclusion

The developed system demonstrates a complete machine learning pipeline for sustainable infrastructure monitoring. Candidate Elimination and ID3 provide symbolic learning and hypothesis-space analysis, while Perceptron and MLP provide connectionist learning capabilities. EM and Bayesian Belief Networks address missing information and probabilistic reasoning. LWR, Naïve Bayes, and Genetic Algorithm are combined to form an optimized hybrid framework.

The experimental benchmarking provides a quantitative comparison of runtime, memory consumption, accuracy, and convergence. The final framework demonstrates how machine learning optimization can support computationally efficient and sustainable infrastructure systems in accordance with SDG Target 9.4.

CODE:

```
import os, time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import Perceptron
from sklearn.neural_network import MLPClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

np.random.seed(42)
os.makedirs("results/figures", exist_ok=True)

SIZES = [1000, 10000, 100000]
features = ["lat", "lon", "temp", "humidity", "pressure",
            "vibration", "noise", "stress", "power", "wind"]

# ----- DATASET -----
def make_data(n):
    x = np.random.randn(n, 10)
    x[:, 0] = np.random.uniform(8, 37, n)
    x[:, 1] = np.random.uniform(68, 97, n)
    x[:, 2] = np.random.normal(30, 7, n)
    x[:, 3] = np.random.uniform(25, 95, n)
    x[:, 4] = np.random.normal(1013, 15, n)
    x[:, 5] = np.random.exponential(2, n)
    x[:, 6] = np.random.normal(50, 12, n)
    x[:, 7] = np.random.normal(30, 8, n)
    x[:, 8] = np.random.normal(500, 100, n)
    x[:, 9] = np.random.uniform(0, 30, n)

    score = .05*x[:, 2] + .8*x[:, 5] + .03*x[:, 6] + .04*x[:, 7]
```

```
y = (score > np.percentile(score,75)).astype(int)
```

```
return pd.DataFrame(x,columns=features), y
```

```
# ----- CANDIDATE ELIMINATION -----
```

```
def candidate_elimination(X,y):
```

```
    S = ["Ø"]*10
```

```
    G = [["?"]*10]
```

```
    for row,target in zip(X.values,y):
```

```
        if target == 1:
```

```
            for i in range(10):
```

```
                if S[i] == "Ø":
```

```
                    S[i] = row[i]
```

```
                elif S[i] != row[i]:
```

```
                    S[i] = "?"
```

```
            else:
```

```
                G = [g for g in G if
```

```
                    all(g[i]=="?" or g[i]==row[i] for i in range(10))]
```

```
    return S,G
```

```
# ----- ENTROPY -----
```

```
def entropy(y):
```

```
    p = np.bincount(y)/len(y)
```

```
    return -sum(v*np.log2(v) for v in p if v>0)
```

```
# ----- EM -----
```

```
def em(X):
```

```
    a = X[:,2].copy()
```

```
    a[np.random.choice(len(a),int(.05*len(a)),False)] = np.nan
```

```
    mean = np.nanmean(a)
```

```
    history = []
```

```

for _ in range(20):
    a[np.isnan(a)] = mean
    mean = np.mean(a)
    history.append(-np.sum((a-mean)**2))

```

```

return mean,np.var(a),history

```

```

# ----- LWR -----

```

```

def lwr(X,y,test,bw):
    out=[]
    for q in test:
        d=np.linalg.norm(X-q,axis=1)
        w=np.exp(-(d*d)/(2*bw*bw))+1e-9
        out.append(np.sum(w*y)/np.sum(w))
    return np.array(out)

```

```

# ----- MAIN -----

```

```

results=[]
mlp_loss={}
em_loss={}

```

```

for n in SIZES:

```

```

    print("\n"+"="*60)
    print("DATASET SIZE:",n)
    print("="*60)

```

```

    X,y=make_data(n)
    X.to_csv(f"results/dataset_{n}.csv",index=False)

```

```

    Xs=StandardScaler().fit_transform(X)
    Xtr,Xte,ytr,yte=train_test_split(
        Xs,y,test_size=.2,random_state=42,stratify=y)

```

```

    # 1 Candidate Elimination
    print("\n[1] Candidate Elimination")

```



```
S,G=candidate_elimination(  
    X.iloc[:min(n,3000)],y[:min(n,3000)])  
print("S =",S)  
print("G size =",len(G))  
print("Version Space Collapse:",len(G)==0)
```

2 ID3

```
print("\n[2] ID3")  
start=time.perf_counter()  
tree=DecisionTreeClassifier(  
    criterion="entropy",random_state=42)  
tree.fit(Xtr,ytr)  
acc=accuracy_score(yte,tree.predict(Xte))  
rt=(time.perf_counter()-start)*1000  
print("Entropy =",round(entropy(y),4))  
print("Accuracy =",round(acc,4))  
results.append([n,"ID3",rt,acc,n*10])
```

3 Perceptron

```
print("\n[3] Perceptron")  
start=time.perf_counter()  
p=Perceptron(max_iter=100,random_state=42)  
p.fit(Xtr,ytr)  
acc=accuracy_score(yte,p.predict(Xte))  
rt=(time.perf_counter()-start)*1000  
print("Accuracy =",round(acc,4))  
results.append([n,"Perceptron",rt,acc,n*10])
```

4 MLP

```
print("\n[4] MLP + Backpropagation")  
start=time.perf_counter()  
mlp=MLPClassifier(  
    hidden_layer_sizes=(16,8),  
    max_iter=150,random_state=42)  
mlp.fit(Xtr,ytr)  
acc=accuracy_score(yte,mlp.predict(Xte))  
rt=(time.perf_counter()-start)*1000  
mlp_loss[n]=mlp.loss_curve_
```

```
print("Accuracy =",round(acc,4))
print("Iterations =",len(mlp.loss_curve_))
results.append([n,"MLP",rt,acc,n*150])
```

5 EM

```
print("\n[5] EM Algorithm")
mean,var,h=em(X.values)
em_loss[n]=h
print("Estimated Mean =",round(mean,4))
print("Estimated Variance =",round(var,4))
```

6 BBN

```
print("\n[6] Bayesian Belief Network")
temp=X.temp > X.temp.median()
vib=X.vibration > X.vibration.median()
bbn=pd.DataFrame({
    "temp":temp,"vibration":vib,"anomaly":y
}).groupby(["temp","vibration"])["anomaly"].mean()
print("Conditional probabilities:")
print(bbn)
```

7 MDL

```
print("\n[7] MDL")
prob=mlp.predict_proba(Xte)
prob=prob[np.arange(len(yte)),yte]
data_cost=-np.sum(np.log2(np.clip(prob,1e-10,1)))
```

```
params=sum(w.size for w in mlp.coefs_)+\
    sum(b.size for b in mlp.intercepts_)
```

```
mdl=data_cost+(params/2)*np.log2(len(yte))
print("MLP Parameters =",params)
print("MLP MDL =",round(mdl,2))
```

8 Naive Bayes

```
print("\n[8] Naive Bayes")
start=time.perf_counter()
nb=GaussianNB().fit(Xtr,ytr)
```

```

acc=accuracy_score(yte,nb.predict(Xte))
rt=(time.perf_counter()-start)*1000
print("Accuracy =",round(acc,4))
results.append([n,"Naive Bayes",rt,acc,n*10])

```

9 GA

```

print("\n[9] Genetic Algorithm")
best_bw=1
best_fit=0
ga_history=[]

```

```

for _ in range(8):
    bw=np.random.uniform(.3,2)
    pred=lwr(
        Xtr[:300],ytr[:300],
        Xte[:100],bw)
    fit=accuracy_score(
        yte[:100],(pred>.5).astype(int))

    if fit>best_fit:
        best_fit=fit
        best_bw=bw
    ga_history.append(best_fit)

```

```

print("Best Bandwidth =",round(best_bw,4))
print("Best Fitness =",round(best_fit,4))

```

10 Hybrid

```

print("\n[10] Hybrid LWR + Naive Bayes + GA")
start=time.perf_counter()

```

```

train_lwr=lwr(Xtr[:300],ytr[:300],Xtr[:300],best_bw)
test_lwr=lwr(Xtr[:300],ytr[:300],Xte,best_bw)

```

```

hnb=GaussianNB().fit(
    train_lwr.reshape(-1,1),ytr[:300])

```

```

acc=accuracy_score(

```

```

yte,hnb.predict(test_lwr.reshape(-1,1)))

rt=(time.perf_counter()-start)*1000

print("Hybrid Accuracy =",round(acc,4))
results.append([n,"Hybrid",rt,acc,n*300])

# ----- SAVE RESULTS -----
df=pd.DataFrame(
    results,
    columns=["Dataset_Size","Algorithm",
             "Runtime_ms","Accuracy","Operations"])

df.to_csv(
    "results/benchmark_results.csv",
    index=False)

# ----- GRAPH FUNCTION -----
def graph(column,title,ylabel,file):

    plt.figure(figsize=(8,5))

    for alg in df.Algorithm.unique():
        d=df[df.Algorithm==alg]
        plt.plot(
            d.Dataset_Size,
            d[column],
            marker="o",
            label=alg)

    plt.xlabel("Dataset Size")
    plt.ylabel(ylabel)
    plt.title(title)
    plt.legend()
    plt.grid()
    plt.tight_layout()

```

```
plt.savefig(  
    f"results/figures/{file}",  
    dpi=300)  
plt.close()
```

1 Runtime

```
graph(  
    "Runtime_ms",  
    "Runtime vs Dataset Size",  
    "Runtime (ms)",  
    "1_Runtime.png")
```

2 Operations

```
graph(  
    "Operations",  
    "Operation Count vs Dataset Size",  
    "Operations",  
    "2_Operations.png")
```

3 Memory

```
memory=[]  
  
for n in SIZES:  
    X,_=make_data(n)  
    memory.append(  
        X.memory_usage().sum()/1024**2)  
  
plt.figure(figsize=(8,5))  
plt.plot(SIZES,memory,marker="o")  
plt.xlabel("Dataset Size")  
plt.ylabel("Memory (MB)")  
plt.title("Memory Usage vs Dataset Size")  
plt.grid()  
plt.tight_layout()  
plt.savefig(  
    "results/figures/3_Memory.png",  
    dpi=300)
```

```
plt.close()
```

```
# 4 Accuracy
```

```
graph(  
    "Accuracy",  
    "Accuracy vs Dataset Size",  
    "Accuracy",  
    "4_Accuracy.png")
```

```
# 5 MLP convergence
```

```
plt.figure(figsize=(8,5))
```

```
for n,v in mlp_loss.items():  
    plt.plot(v,label=str(n))
```

```
plt.xlabel("Iteration")  
plt.ylabel("Loss")  
plt.title("MLP Error Convergence")  
plt.legend(title="Records")  
plt.grid()  
plt.tight_layout()  
plt.savefig(  
    "results/figures/5_MLP_Convergence.png",  
    dpi=300)  
plt.close()
```

```
# 6 EM convergence
```

```
plt.figure(figsize=(8,5))
```

```
for n,v in em_loss.items():  
    plt.plot(v,label=str(n))
```

```
plt.xlabel("Iteration")  
plt.ylabel("Error")  
plt.title("EM Convergence")  
plt.legend(title="Records")  
plt.grid()  
plt.tight_layout()
```

```
plt.savefig(  
    "results/figures/6_EM_Convergence.png",  
    dpi=300)  
plt.close()
```

```
print("\n"+"="*60)  
print("EXPERIMENT COMPLETED SUCCESSFULLY")  
print("="*60)  
print(df.to_string(index=False))  
print("\n6 graphs saved in: results/figures/")
```

RESULTS:

